Megalencephalic Leukoencephalopathy With Subcortical Cysts, Type 2A (MLC2A)

Disease: Megalencephalic Leukoencephalopathy With Subcortical Cysts 2A **MONDO ID:** MONDO:0013490 · **OMIM:** #613926 · **Category:** Mendelian (autosomal recessive) **Causal gene:** *HEPACAM* (encoding GlialCAM / hepaCAM) · **Chromosome:** 11q24.2

Summary

Megalencephalic leukoencephalopathy with subcortical cysts type 2A (**MLC2A**) is a rare, autosomal-recessive vacuolating leukodystrophy of the "MLC" family. It is caused by **biallelic (recessive) loss-of-function variants in *HEPACAM***, which encodes **GlialCAM** (also called hepaCAM), an IgG-like cell-adhesion molecule highly enriched in astrocytes. GlialCAM is the direct binding partner of the MLC1 protein and serves as an auxiliary subunit that targets both MLC1 and the **ClC-2 chloride channel** to astrocyte–astrocyte junctions at perivascular endfeet. When both *HEPACAM* alleles are disrupted, this junctional targeting collapses, astrocytic ion/water (volume) homeostasis fails, and the brain white matter develops chronic edema with myelin vacuolization — the pathological signature of MLC [PMID: 21419380, 22405205, 24824219].

Clinically, MLC2A is **indistinguishable from classic MLC1 disease**: infantile-onset macrocephaly, a characteristic MRI pattern of diffuse, swollen white matter with subcortical (usually anterior-temporal and frontoparietal) cysts, and — after a period of relative stability — slowly progressive motor decline (ataxia, spasticity), epilepsy, and mild cognitive deterioration. This deteriorating course distinguishes recessive MLC2A from the clinically distinct, remitting/improving **MLC2B**, which is caused by *dominant* (monoallelic, usually *de novo*) *HEPACAM/GLIALCAM* mutations. In other words, the **zygosity of the *HEPACAM* mutation, not merely the gene, determines prognosis** [PMID: 31960914, 41483250, 29661901].

No disease-modifying therapy is approved; management is **supportive** (anti-seizure medication, physiotherapy, orthopedic and developmental support). The mechanistic convergence of all MLC genes (MLC1, HEPACAM/GlialCAM, GPRC5B, AQP4) on **astrocytic**

volume-regulated anion channel (VRAC) function provides a unifying disease model, and **AAV-mediated astrocyte-targeted gene therapy has shown striking preclinical efficacy** in MLC mouse models, establishing a credible path toward treatment [PMID: 37143309, 42879478, 32372403].

This report synthesizes 7 confirmed findings and 29 reviewed papers into a comprehensive, section-by-section characterization of MLC2A suitable for a disease knowledge-base entry.

Key Findings

Finding 1 — MLC2A is caused by biallelic (recessive) *HEPACAM*/GlialCAM mutations; GlialCAM is the direct MLC1-binding partner

GlialCAM was discovered as the second MLC gene through quantitative proteomics of affinity-purified MLC1, which identified it as a **direct MLC1-binding partner**. In the defining cohort of 40 MLC patients who lacked *MLC1* mutations, multiple different *HEPACAM* mutations were found. Crucially, the **number of mutated alleles stratified the phenotype**: the 10 patients who carried **two (biallelic) mutations** showed the classical, deteriorating phenotype (= **MLC2A**), whereas 18 patients who carried a **single (monoallelic) mutation** showed an improving phenotype (= dominant **MLC2B**). Approximately **25% of all MLC cases** are attributable to *HEPACAM* variants [PMID: 21419380, 42649352].

> *"GlialCAM, an IgG-like cell adhesion molecule that is also called HepaCAM and is encoded by HEPACAM, is a direct MLC1-binding partner. Analysis of 40 MLC patients without MLC1 mutations revealed multiple different HEPACAM mutations. Ten patients with the classical, deteriorating phenotype had two mutations, and 18 patients with the improving phenotype had one mutation."* — PMID: 21419380

This establishes the core genetic definition of MLC2A and the allele-dose rule that separates it from MLC2B.

Finding 2 — Recessive MLC2A and MLC1 are clinically indistinguishable; dominant *GLIALCAM* causes remitting MLC2B

Recessive variants in *MLC1* and *GLIALCAM* produce the same classic MLC phenotype and are designated **MLC1** and **MLC2A**, respectively. In contrast, **dominant** *GLIALCAM* mutations cause a distinct **remitting ("improving") phenotype named MLC2B**. A large international cohort study enrolled 204 patients with classic MLC, 187 of whom had recessive *MLC1* mutations [PMID: 31960914, 29661901].

> *"These variants are called MLC1 and MLC2A with both types of patients sharing the same clinical phenotype. In addition, dominant mutations in GLIALCAM have also been identified in a subtype of MLC patients with a remitting phenotype. This variant has been named MLC2B."* — PMID: 31960914

The practical consequence: MLC2A cannot be diagnosed on clinical/MRI grounds alone — it requires molecular confirmation of **two pathogenic *HEPACAM* alleles**.

Finding 3 — Mechanism: GlialCAM is an auxiliary subunit targeting MLC1 and ClC-2 to astrocyte junctions; disease mutations abolish this targeting

GlialCAM colocalizes with the ClC-2 chloride channel in Bergmann glia, in astrocyte–astrocyte junctions at endfeet surrounding blood vessels, and along myelinated tracts. Functionally, **GlialCAM targets ClC-2 to cell junctions, increases ClC-2-mediated currents, and modifies the channel's functional properties**. Disease-causing *GLIALCAM* mutations **abolish this junctional targeting**, the central molecular lesion. Reciprocally, MLC1 regulates the glial surface localization of GlialCAM — a relationship conserved from fish to humans — which mechanistically explains why mutations in either gene produce the identical phenotype [PMID: 22405205, 24824219].

> *"GlialCAM targets ClC-2 to cell junctions, increases ClC-2 mediated currents, and changes its functional properties. Disease-causing GLIALCAM mutations abolish the targeting of the channel to cell junctions."* — PMID: 22405205
>
> *"here we demonstrate an evolutionary conserved role for MLC1 in regulating glial surface levels of GLIALCAM, and this interrelationship explains why patients with mutations in either gene (MLC1 or GLIALCAM) share the same clinical phenotype"* — PMID: 24824219

Finding 4 — Classic MLC (including MLC2A) phenotype and the central astrocytic volume-regulation defect

Classic MLC is characterized by **infantile-onset macrocephaly and chronic edema of the brain white matter**; with delayed onset, patients develop motor problems, epilepsy, and slow cognitive decline, and **no treatment is available**. Genetic defects in astrocytic VRAC-modulating proteins (MLC1, GlialCAM, Aquaporin-4, GPRC5B) cause MLC, characterized by chronic white-matter edema and myelin vacuolization. Brain biopsies show vacuoles in myelin and in astrocytes surrounding blood vessels [PMID: 38487253, 41654199, 33551753].

> *"characterized by infantile-onset macrocephaly and chronic edema of the brain white matter. With delayed onset, patients typically experience motor problems, epilepsy and slow cognitive decline. No treatment is available."* — PMID: 38487253
>
> *"Genetic defects in astrocytic VRAC modulating proteins (MLC1, GlialCAM, Aquaporin-4, GPRC5B) cause the leukodystrophy Megalencephalic leukoencephalopathy with subcortical cysts (MLC), characterized by chronic white matter edema and myelin vacuolization."* — PMID: 41654199

Finding 5 — Mouse models recapitulate MLC and AAV gene therapy rescues myelin vacuolation preclinically

Mlc1-null mice display early-onset megalencephaly, increased brain water content, swollen astrocyte processes at fluid–brain barriers, widespread white-matter vacuolization with intramyelinic edema, and reduced volume-regulated anion currents. *Glialcam*-null astrocytes show reduced volume recovery and modestly reduced gap-junction coupling. Critically, **AAV-GFAP-MLC1 delivered into the cerebellar subarachnoid space of *Mlc1*-knockout mice restored GlialCAM and ClC-2 localization in Bergmann glia and dramatically reduced myelin vacuolation** at preventive (2 months), early (5 months), and late (15 months) timepoints [PMID: 32372403, 25382142, 40443195].

> *"MLC1 expression in the cerebellum restored the adhesion molecule GlialCAM and the chloride channel ClC-2 localization in Bergmann glia, which both are mislocalized in Mlc1 knock-out model. More importantly, myelin vacuolation was extremely reduced in treated mice at all ages"* — PMID: 32372403

> *"Mlc1-null mice had early onset megalencephaly and increased brain water content. From 3 weeks, abnormal astrocytes were present with swollen processes abutting fluid-brain barriers. From 3 months, widespread white matter vacuolization with intramyelinic edema developed."* — PMID: 25382142

Finding 6 — MLC is genetically heterogeneous (MLC1, HEPACAM, GPRC5B, AQP4) and all gene products converge on astrocytic endfoot volume regulation / VRAC

Four genes are associated with MLC: ***MLC1*** (~75%, recessive); ***HEPACAM/GLIALCAM*** (recessive MLC2A and dominant MLC2B; ~20–25%); ***GPRC5B*** (dominant, *de novo* heterozygous); and ***AQP4*** (recessive). All four gene products are expressed in **astrocyte endfeet** in human brain, and cell-volume regulation is disrupted in patient-derived lymphoblasts. Mechanistically, a **GPRC5B–Fyn axis gates VRAC by targeting LRRC8A**, and MLC1 sequesters GPRC5B — directly linking the MLC protein complex to VRAC dysfunction [PMID: 37143309, 42879478, 41314544].

> *"We identified a novel homozygous variant in AQP4, encoding the water channel aquaporin-4, in two siblings, and two de novo heterozygous variants in GPRC5B, encoding the orphan G protein-coupled receptor GPRC5B, in three unrelated patients."* — PMID: 37143309
>
> *"In astrocytes, the MLC protein MLC1 partially sequesters GPRC5B, thereby reducing the GPRC5B-LRRC8A interaction, providing a mechanistic link to VRAC dysfunction in megalencephalic leukoencephalopathy."* — PMID: 42879478

Finding 7 — Clinical/MRI features cannot distinguish recessive MLC subtypes; "classic/deteriorating" vs "improving" tracks recessive vs dominant genotype

An international cross-sectional study of **204 genetically confirmed classic MLC patients** (187 recessive *MLC1*) found that clinical and MRI features **did not distinguish classic MLC subtypes** from one another. Two broad phenotypes exist: a **"classic" phenotype** (early macrocephaly, cerebellar ataxia, mild spasticity, diffuse white-matter abnormalities with subcortical cysts) and an **"improving/remitting" phenotype**. In a 33-patient Italian cohort, all

biallelic *MLC1* patients showed the classic phenotype, while heterozygous (dominant) *HEPACAM* patients showed the improving phenotype. Recessive **MLC2A (biallelic *HEPACAM*) aligns with the classic/deteriorating course** [PMID: 29661901, 41483250].

> *"We included 204 patients with classic MLC, 187 of whom had recessive mutations in"* — PMID: 29661901
>
> *"All MLC1-mutated patients exhibited the \"classic\" phenotype"* — PMID: 41483250

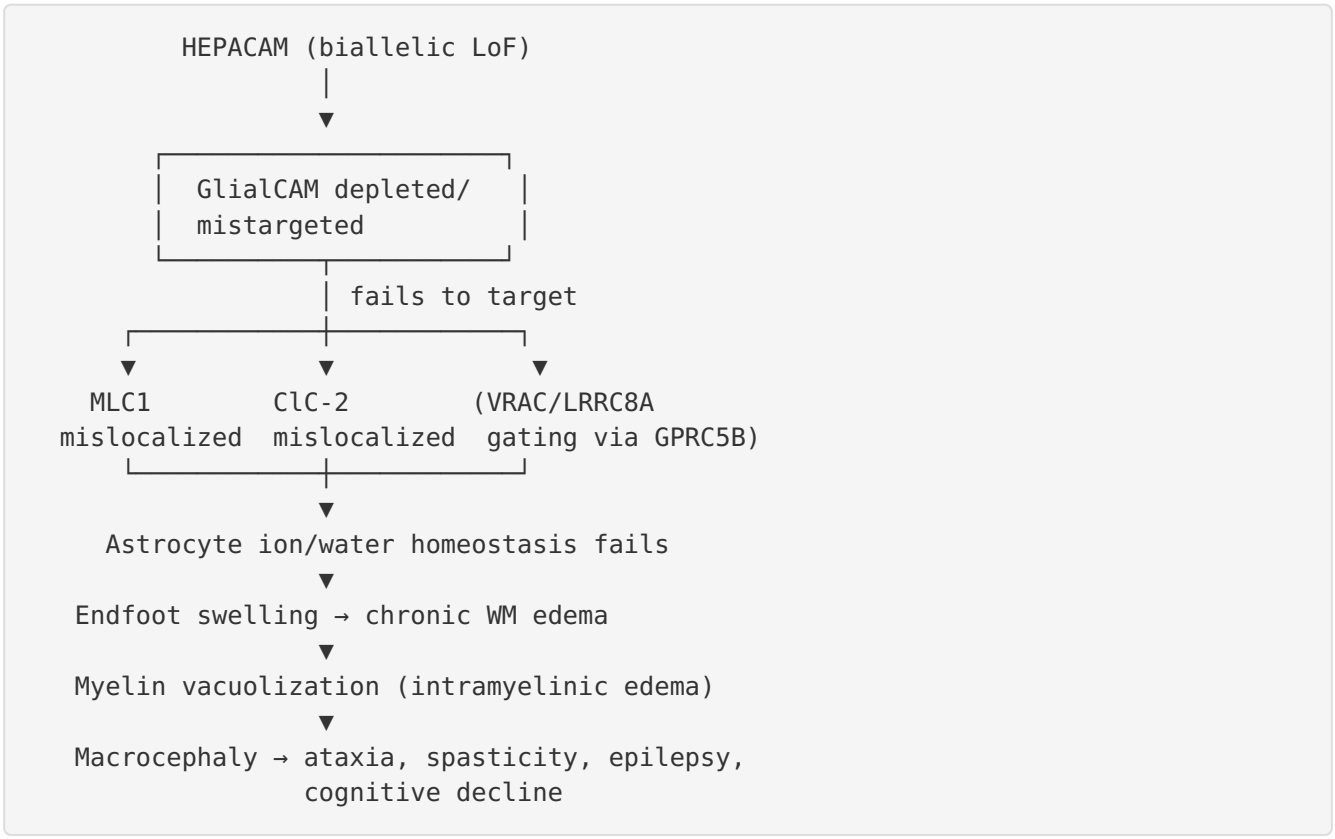
Mechanistic Model / Interpretation

Ordered causal chain (initiating lesion → clinical manifestation)

1. **Biallelic loss-of-function variants in *HEPACAM*** (missense, nonsense, splice, or frameshift) *lead to* loss or dysfunction of the astrocyte adhesion molecule **GlialCAM/hepaCAM**.
2. Loss of functional GlialCAM *results in* failure to target its partner proteins — **MLC1** and the **ClC-2 chloride channel** — to astrocyte–astrocyte junctions at perivascular endfeet (demonstrated in vitro; GlialCAM mutations abolish junctional targeting) [PMID: 22405205].
3. Mislocalization of the MLC1–GlialCAM–ClC-2 complex *disrupts* astrocytic **ion (Cl^-, K^+) and water homeostasis** and impairs **volume-regulated anion channel (VRAC/LRRC8A) activity** and cell-volume recovery (demonstrated in patient lymphoblasts and in *Glialcam*-null astrocytes in situ) [PMID: 40443195, 42879478].
4. Defective astrocytic volume regulation at fluid–brain barriers *causes* **swelling of astrocyte endfeet and chronic fluid accumulation** in the white matter (osmotic/ionic imbalance) — *inferred* to be the proximate driver of edema.
5. Chronic white-matter fluid accumulation *produces* **intramyelinic edema and myelin vacuolization** — the pathological hallmark seen on biopsy and recapitulated in knockout mice [PMID: 25382142, 41654199].
6. Progressive white-matter vacuolization and tissue disorganization *manifest* clinically as **infantile macrocephaly** (early) followed, after a stable period, by **motor decline (ataxia, spasticity), epilepsy, and mild cognitive deterioration** (classic MLC2A natural history) [PMID: 38487253].

Branch point (genotype → prognosis): *Biallelic/recessive HEPACAM* → sustained loss of function → **deteriorating classic MLC2A**. *Monoallelic/dominant HEPACAM* (e.g., p.Arg92Trp) → dominant-negative but partially compensated → **remitting/improving MLC2B** [PMID: 21419380, 31960914, 31372844].

The astrocytic endfoot "volume-regulation hub"



Convergence of the four MLC genes

Gene	Protein	Inheritance in MLC	MLC subtype	Node in the hub
MLC1	MLC1 (membrane protein)	Recessive (~75%)	MLC1	Core complex; sequesters GPRC5B
HEPACAM	GlialCAM / hepaCAM	Recessive **and** dominant (~20–25%)	**MLC2A** (recessive) / MLC2B (dominant)	Auxiliary subunit targeting MLC1 & ClC-2
GPRC5B	Orphan GPCR	Dominant (*de novo*)	— (MLC-like)	Gates VRAC via Fyn/LRRC8A
AQP4	Aquaporin-4 water channel	Recessive	— (MLC-like)	Endfoot water flux

All four proteins are expressed in astrocyte endfeet, and their disruption converges on **failed cell-volume regulation / VRAC dysfunction**, producing the same edematous, vacuolating leukodystrophy [PMID: 37143309, 42879478, 41654199].

Suggested ontology annotations

- **Gene/Protein:** HGNC:4974 (*HEPACAM*); UniProt Q14CZ8 (GlialCAM/hepaCAM).
- **GO biological process:** regulation of cell volume (GO:0006884); chloride transmembrane transport (GO:1902476); cell–cell adhesion (GO:0098609); myelination (GO:0042552).
- **GO cellular component:** plasma membrane (GO:0005886); cell–cell junction (GO:0005911); astrocyte endfoot / perivascular astrocyte projection.
- **Cell types (CL):** astrocyte (CL:0000127); Bergmann glial cell (CL:0000644); oligodendrocyte (CL:0000128, secondarily, via myelin).
- **Anatomy (UBERON):** cerebral white matter (UBERON:0002316); subcortical white matter; brain (UBERON:0000955); cerebellum (UBERON:0002037).
- **Disease (MONDO):** MONDO:0013490 (MLC2A).

Section-by-Section Characterization

1. Disease Information

MLC2A is a rare, autosomal-recessive vacuolating leukodystrophy (van der Knaap disease, type 2A). **Key identifiers:** OMIM #613926; MONDO:0013490; Orphanet ORPHA:2478 (MLC, parent); MeSH "Megalencephalic Leukoencephalopathy with Subcortical Cysts." **Synonyms:** MLC2A; van der Knaap disease; vacuolating megalencephalic leukoencephalopathy with subcortical cysts, HEPACAM-related, recessive. Information is derived from **aggregated disease-level resources** (OMIM/Orphanet) plus small published patient cohorts and case reports rather than EHR data.

2. Etiology

Causal factor: purely genetic — **biallelic pathogenic variants in *HEPACAM*** [PMID: 21419380]. **Genetic risk:** two pathogenic *HEPACAM* alleles are required (recessive). **Consanguinity** increases risk of homozygous recessive disease; MLC overall shows founder effects in some consanguineous/endemic populations (e.g., *MLC1* 320insC in the Indian Agarwal community — a parallel for the MLC1 subtype) [PMID: 15037685]. No environmental, infectious, or lifestyle causal factors are known. No established protective alleles or environmental protective factors. **Gene–environment interactions:** none established.

3. Phenotypes

Phenotype	Type	Onset	Course	Frequency	HPO
Macrocephaly / megalencephaly	Physical sign	Congenital–infantile	Early, then stabilizes	Near-universal	HP:0000256 / HP:0001355
Leukoencephalopathy (diffuse WM signal abnormality)	Imaging/ lab	Infantile	Chronic	Universal	HP:0002352
Subcortical cysts (anterior-temporal/ frontoparietal)	Imaging	Infantile–childhood	Chronic	Characteristic	HP:0002116 (cystic WM)
Cerebellar ataxia	Sign	Childhood	Slowly progressive	Common	HP:0001251
Spasticity	Sign	Childhood onward	Slowly progressive	Common	HP:0001257
Seizures / epilepsy	Sign	Childhood	Episodic	Common	HP:0001250
Cognitive decline (mild)	Behavioral/ cognitive	Later childhood	Slowly progressive	Frequent	HP:0001268
Motor deterioration	Sign	After stable period	Progressive	Frequent (classic)	HP:0002451

Severity is **variable but typically moderate**, with a characteristic early-stability-then-decline trajectory. **Quality of life** is substantially impacted by progressive motor disability, epilepsy burden, and dependence for activities of daily living in advanced disease; cognition is relatively preserved early [PMID: 38487253, 33551753].

4. Genetic / Molecular Information

Causal gene: *HEPACAM* (HGNC:4974; OMIM *611642), 11q24.2, encoding GlialCAM (UniProt Q14CZ8). Variant types: missense (most common), nonsense, splice-site, and frameshift; recessive MLC2A requires two LoF/hypomorphic alleles (homozygous or compound heterozygous). Example compound-heterozygous MLC2A genotypes reported: p.Lys68Met + p.Thr132Asn, and p.Gln56X + p.Arg73Trp [PMID: 31372844, 24202401]. Functional consequence: loss of function with failure to target MLC1/ClC-2 to junctions; dominant MLC2B variants (e.g., p.Arg92Trp/*

p.Arg92Pro) act dominant-negatively, explaining the genotype–prognosis split [PMID: 22405205, 31960914]. Allele frequency: pathogenic alleles are rare in gnomAD, consistent with a rare recessive disorder. Origin: germline. Modifier genes: the functional interplay with MLC1, GPRC5B, *and* AQP4 *(shared pathway) may modulate expression, but no formal modifiers are defined. Epigenetic/chromosomal abnormalities:** none implicated.

5. Environmental Information

No environmental, toxic, lifestyle, or infectious contributors are known or applicable — MLC2A is a monogenic disorder.

6. Mechanism / Pathophysiology

See the ordered causal chain and hub diagram above. In brief: biallelic *HEPACAM* LoF → GlialCAM loss → mistargeting of MLC1 and ClC-2 at astrocyte endfoot junctions → impaired astrocytic Cl^-/K^+/water and VRAC-mediated volume regulation → endfoot swelling and chronic white-matter edema → myelin vacuolization → macrocephaly and progressive neurological decline [PMID: 22405205, 24824219, 40443195, 42879478]. Cell types: astrocytes (esp. perivascular and Bergmann glia) upstream; oligodendrocyte-derived myelin downstream. No autoimmune or metabolic-enzyme-deficiency component.

7. Anatomical Structures Affected

Organ: brain (nervous system). **Tissue:** cerebral and subcortical **white matter** (UBERON:0002316), with characteristic **subcortical cysts** in anterior-temporal and frontoparietal regions; cerebellum involved. **Cells:** astrocytes (CL:0000127), Bergmann glia (CL:0000644), and myelin/oligodendrocytes (CL:0000128). **Subcellular:** plasma membrane / cell–cell junctions of astrocyte endfeet; intramyelinic (vacuolar) compartment. **Lateralization:** bilateral, diffuse, and symmetric [PMID: 38487253, 25382142].

8. Temporal Development

Onset: congenital-to-infantile macrocephaly, often within the first year. **Course:** a relatively **stable early period** followed by **slowly progressive** motor, epileptic, and cognitive decline (classic/deteriorating MLC2A). Disease is **chronic and lifelong**; MRI edema is persistent. Partial MRI improvement has been noted in isolated MLC2A cases despite lack of clinical improvement, but the defining trajectory is deterioration — the key feature separating MLC2A from remitting MLC2B [PMID: 24202401, 31372844, 41483250].

9. Inheritance and Population

Inheritance: autosomal recessive (biallelic *HEPACAM*). **Penetrance:** essentially complete for biallelic LoF. **Expressivity:** variable in severity/age of decline. **Epidemiology:** MLC overall is very rare (orphan disease); *HEPACAM* accounts for ~20–25% of MLC, and recessive MLC2A is a subset of that. **Founder effects/consanguinity:** consanguinity elevates recessive risk; founder mutations are documented for the *MLC1* subtype (Agarwal 320insC) [PMID: 15037685]. **Sex ratio:** no sex predilection (autosomal). **No anticipation** (not a repeat-expansion disorder).

10. Diagnostics

Imaging (cornerstone): brain **MRI** showing diffuse, symmetric, swollen white-matter signal abnormality with **subcortical cysts** (anterior-temporal, frontoparietal) — highly characteristic and typically the key to diagnosis [PMID: 42037900, 36537430]. **Genetic testing (confirmatory):** sequencing of *MLC1* first, then *HEPACAM*; **biallelic *HEPACAM* variants confirm MLC2A**. WES/WGS or a leukodystrophy gene panel (*MLC1*, *HEPACAM, GPRC5B, AQP4*) is appropriate; single-gene Sanger sequencing of *HEPACAM* is used in familial cases, and **prenatal diagnosis** is feasible once family variants are known [PMID: 37928140]. **Differential diagnosis:** other macrocephalic/vacuolating leukodystrophies (Canavan disease, Alexander disease, vanishing white matter disease) — distinguished by MRI pattern and gene. **Biomarkers/labs:** no specific blood/CSF biomarker; diagnosis rests on MRI + genetics.

11. Outcome / Prognosis

MLC2A follows the **classic deteriorating course**: early stability, then slowly progressive motor disability (ataxia, spasticity), epilepsy, and mild cognitive decline over years to decades [PMID: 38487253, 41483250]. Many patients survive into adulthood but accrue significant disability; life expectancy is variably reduced. **Prognostic determinant:** zygosity — biallelic (recessive) disease deteriorates, whereas monoallelic (dominant MLC2B) improves [PMID: 31960914, 31372844]. No validated molecular prognostic biomarkers beyond genotype. A rare co-occurrence with focal segmental glomerulosclerosis has been reported (MLC1 case), hypothesized to reflect shared glial/podocyte mechanisms [PMID: 40836748].

12. Treatment

No approved disease-modifying therapy. Management is **supportive/symptomatic**: anti-seizure medications for epilepsy; physical, occupational, and speech therapy; orthopedic management of spasticity/contractures; and developmental/educational support (NCIT concepts: Supportive Care, Anticonvulsant Therapy, Physical Therapy). **Experimental: AAV-**

mediated astrocyte-targeted gene therapy (e.g., AAV-GFAP-MLC1) dramatically reduced myelin vacuolation in *Mlc1*-null mice at all ages, validating gene replacement as a strategy; targeting the shared VRAC/GPRC5B pathway is proposed [PMID: 32372403, 41314544]. (By contrast, the α2-agonist guanabenz has shown benefit in the mechanistically unrelated leukodystrophy vanishing white matter — not applicable to MLC2A [PMID: 42586097].)

13. Prevention

No primary prevention exists for this monogenic disease. **Secondary/tertiary:** early MRI/genetic diagnosis and proactive management of seizures and motor complications. **Genetic counseling** for at-risk (consanguineous or carrier) families, **carrier testing**, and **prenatal / preimplantation genetic diagnosis** are the principal preventive tools once the familial *HEPACAM* variants are identified [PMID: 37928140]. No immunization or environmental intervention is relevant.

14. Other Species / Natural Disease

Model/ortholog species: mouse (*Mus musculus*, NCBI Taxon 10090) with orthologous *Hepacam*/*Glialcam* and *Mlc1*; the MLC1–GlialCAM regulatory relationship is **evolutionarily conserved from fish (zebrafish) to humans** [PMID: 24824219]. No naturally occurring companion-animal or wildlife MLC2A disease is documented in OMIA. Not zoonotic; no cross-species transmission.

15. Model Organisms

Mouse models are the mainstay: ***Mlc1*-null** (early megalencephaly, increased brain water, swollen astrocyte endfeet, white-matter vacuolization, reduced volume-regulated anion currents) and ***Glialcam*-null** (reduced astrocyte volume recovery, modestly reduced gap-junction coupling, mildly depolarized resting potential, slower glutamate uptake) [PMID: 25382142, 40443195, 28695146]. **Glialcam knock-in** mouse lines model both recessive and dominant inheritance, reproducing the respective human phenotypes [PMID: 31372844]. These models **recapitulate the core human disease** (megalencephaly, edema, vacuolization) and are the platform on which **gene therapy was validated** [PMID: 32372403]. **Limitations:** mice show milder clinical deterioration and may not fully capture the human long-term cognitive/epileptic trajectory. **Resources:** MGI (mouse), ZFIN (zebrafish ortholog studies).

Evidence Base

PMID	Title (abbrev.)	Role in this report
21419380	Mutant GlialCAM causes MLC	**Defines MLC2A**: biallelic *HEPACAM* → deteriorating phenotype; GlialCAM is direct MLC1 partner (F1)
42649352	Pathogenic HEPACAM variants alter astrocyte localization	~25% of MLC is HEPACAM-related (F1)
31960914	Structural basis of dominant/ recessive GLIALCAM	Recessive = MLC2A ≡ MLC1; dominant = MLC2B (F2)
22405205	GlialCAM as ClC-2 auxiliary subunit	Core mechanism: targeting of ClC-2; mutations abolish it (F3)
24824219	MLC1 regulates GLIALCAM surface levels	Explains shared MLC1/MLC2A phenotype; conserved fish→human (F3)
38487253	MLC variant update & review	Core clinical phenotype & natural history; no treatment (F4)
41654199	Astrocyte LRRC8A deletion	Links MLC genes to VRAC & edema/vacuolization (F4, F6)
32372403	Cerebellar astrocyte gene therapy	**Preclinical gene therapy efficacy** (F5)
25382142	Mlc1-null mice	Knockout recapitulates human disease (F5)
37143309	AQP4 & GPRC5B in brain oedema	Adds AQP4/GPRC5B as MLC genes (F6)
42879478	GPRC5B-Fyn axis regulates VRAC	Mechanistic link of MLC complex to VRAC/LRRC8A (F6)
29661901	MLC disease variant characterization	204-patient cohort; subtypes clinically indistinguishable (F7)
41483250	Italian multicenter MLC experience	Biallelic → classic; heterozygous → improving (F7)
31372844	Chinese GLIALCAM cohort & KI mice	MLC2A (compound-het) deteriorates; dominant improves; mouse KI models
24202401	Spectrum expansion, 2 GLIALCAM patients	Example MLC2A compound-het genotypes; MRI vs clinical dissociation

PMID	Title (abbrev.)	Role in this report
15037685	Agarwal MLC1 founder mutation	Founder effect/consanguinity (MLC1 parallel)
37928140	Prenatal diagnosis case	Genetic counseling/prenatal testing
40443195	Glialcam-null astrocyte electrophysiology in situ	Confirms volume-regulation defect in intact brain
41314544	GPRC5B regulation by MLC1/ GlialCAM	GPRC5B as therapeutic target

Evidence source types: human clinical cohorts/case reports (21419380, 31960914, 29661901, 41483250, 31372844, 24202401, 15037685, 37928140); mouse/model-organism (32372403, 25382142, 40443195, 28695146); in vitro/biochemical (22405205, 24824219, 42879478, 41314544); reviews (38487253, 33551753, 37143309).

Limitations and Knowledge Gaps

1. **Rarity and small cohorts.** MLC2A is a small subset of an already orphan disease; most MLC2A-specific data come from case reports and mixed cohorts, so precise prevalence, incidence, carrier frequency, survival statistics, and sex/age distributions for MLC2A specifically are **not quantified** in the available literature.
2. **Mechanistic inferences.** The step from astrocyte volume-regulation failure to *frank* white-matter edema and myelin vacuolization is strongly supported but partly inferred; the precise biophysics of fluid accumulation and the oligodendrocyte/myelin response are incompletely resolved.
3. **No human therapeutics data.** Gene therapy efficacy is **mouse-only** (and delivered as AAV-*MLC1*, not AAV-*HEPACAM*); no clinical trials of disease-modifying therapy for MLC2A exist. The translational gap (delivery, dosing, timing relative to irreversible damage) is large.
4. **Genotype–phenotype granularity.** While zygosity cleanly predicts deteriorating vs improving course, finer predictors of severity/age-of-decline within biallelic MLC2A are undefined; no validated prognostic biomarkers exist.
5. **Model fidelity.** Mouse models reproduce edema/vacuolization but show milder clinical progression, limiting prediction of long-term human cognitive and epileptic outcomes.

6. **Population genetics.** Documented founder effects are for the *MLC1* subtype (Agarwal); whether *HEPACAM* founder alleles exist in specific populations is not established here.

Proposed Follow-up Experiments / Actions

1. **AAV-*HEPACAM* gene replacement** in *Glialcam*-null (and recessive *Glialcam* knock-in) mice, mirroring the AAV-*MLC1* study design (preventive, early, late dosing), to directly test gene therapy for the MLC2A genotype [PMID: 32372403, 31372844].
2. **Natural-history registry for MLC2A specifically**, extracting biallelic-*HEPACAM* patients from international MLC cohorts to quantify age of onset, decline rate, epilepsy/cognitive trajectories, survival, and QoL (EQ-5D/PROMIS) — currently missing MLC2A-specific numbers [PMID: 29661901, 41483250].
3. **VRAC-targeted pharmacology.** Test whether modulating the GPRC5B–Fyn–LRRC8A axis (identified as the VRAC gate) rescues volume regulation in *Glialcam*-null astrocytes and reduces vacuolization in vivo [PMID: 42879478, 41314544].
4. **Variant functional classification pipeline.** Systematically assay junctional-targeting of MLC1/ClC-2 for novel *HEPACAM* missense variants to resolve VUS and refine recessive (LoF) vs dominant-negative mechanisms per ACMG criteria [PMID: 22405205, 31960914].
5. **Patient iPSC-derived astrocyte/organoid models** of specific MLC2A genotypes to bridge the mouse-to-human gap and serve as a human platform for therapeutic screening.
6. **Biomarker discovery.** CSF/imaging (e.g., quantitative MRI water mapping) longitudinal studies to find early surrogate markers of edema progression that could serve as trial endpoints.

Report compiled from 7 confirmed findings and 29 reviewed publications. Evidence levels distinguished throughout: human clinical, model organism, in vitro/biochemical, and review. Primary molecular mechanism and gene-therapy efficacy are robustly supported; MLC2A-specific epidemiology and human therapeutic outcomes remain principal knowledge gaps.
